PART II SCHWESER'S QuickSheet

CRITICAL CONCEPTS FOR THE 2025 FRM® EXAM

MARKET RISK MEASUREMENT AND MANAGEMENT

Value at Risk (VaR)

VaR for a given confidence level occurs at the cutoff point that separates the tail losses from the remaining distribution.

Historical simulation approach: order return observations and find the observation that corresponds to the VaR loss level.

Parametric estimation approach: assumes a distribution for the underlying observations.

- Normal distribution assumption:
$$VaR = (-\mu_r + \sigma_r \times z_\alpha)$$
- Lognormal distribution assumption:
$$VaR = (1 - e^{\mu_r - \sigma_r \times z_\alpha})$$

Expected Shortfall (ES)

Estimates tail loss by averaging VaRs for increasing confidence levels in the tail.

Weighted Historical Simulation Approaches

- *Age-weighted*: adjusts the most recent (distant) observations to be more (less) heavily weighted.
- *Volatility-weighted*: replaces historic returns with volatility-adjusted returns; actual procedure of estimating VaR is unchanged.
- *Correlation-weighted*: updates the variance-covariance matrix between assets in the portfolio.
- Filtered historical simulation: relies on bootstrapping of standardized returns based on volatility forecasts; able to capture conditional volatility, volatility clustering, and/or data asymmetry.

Peaks-Over-Threshold (POT)

Application of extreme value theory (EVT) to the distribution of excess losses over a high threshold. The POT approach can be used to compute VaR. From estimates of VaR, we can derive ES.

Backtesting VaR

- Compares the number of instances when losses exceed the VaR level (exceptions or exceedances) with the number predicted by the model at the chosen level of confidence.
- *Failure rate*: number of exceptions divided by number of observations.
- The Basel Committee requires backtesting at the 99% confidence level over one year; establishes zones for the number of exceptions with corresponding penalties (increases in the capital multiplier).

VaR Mapping

Mapping involves finding common risk factors among positions in a given portfolio. It may be difficult and time consuming to manage the risk of each individual position. One can evaluate the value of portfolio positions by mapping them onto common risk factors.

Mean Reversion

- Implies that over time variables or returns regress back to the mean or average return.
- Mean reversion rate, a, is expressed as:
$$S_t - S_{t-1} = a(\mu - S_{t-1})$$
- The β coefficient of a regression is equal to the negative of the mean reversion rate.

Autocorrelation

- Measures the degree that a variable's current value is correlated to past values.
- Has the exact opposite properties of mean reversion.
- The sum of the mean reversion rate and the one-period autocorrelation rate will always equal one.

Correlation Swap

- Used to trade a fixed correlation between two or more assets with a realized correlation.
- Realized correlation for a portfolio of n assets:
$$\rho_{\text{realized}} = \frac{2}{n^2 - n}\sum_{i>j}\rho_{i,j}$$
- Payoff for correlation swap buyer:
notional amount $\times (\rho_{\text{realized}} - \rho_{\text{fixed}})$

Gaussian Copula Model

- Indirectly defines a correlation relationship between two variables.
- Maps the marginal distribution of each variable to a standard normal distribution (done on percentile-to-percentile basis).
- The new joint distribution is a multivariate standard normal distribution.
- A Gaussian default time copula can be used for measuring the joint probability of default between two assets.

Regression-Based Hedge

$$F^R = F^N \times \left(\frac{DV01^N}{DV01^R}\right) \times \beta$$

where:
F^R = face amount of hedging instrument
F^N = face amount of initial position

Bond Valuation Using Binomial Tree

Using backward induction, the value of a bond at a given node in a binomial tree is the average of the present values of the two possible values from the next period. The appropriate discount rate is the forward rate associated with the node under analysis.

There are three basic steps to valuing an option on a fixed-income instrument using a binomial tree.

Step 1: Price the bond value at each node using the projected interest rates.
Step 2: Calculate the intrinsic value of the derivative at each node at maturity.
Step 3: Calculate the expected discounted value of the derivative at each node using the risk-neutral probabilities and work backward through the tree.

Interest Rate Expectations

Expectations play an important role in determining the shape of the yield curve and can be illustrated by examining yield curves that are flat, upward-sloping, and downward-sloping. If expected 1-year spot rates for the next three years are r_1, r_2, and r_3, then the 2-year and 3-year spot rates are computed as follows.

$$\hat{r}(2) = \sqrt{(1 + r_1)(1 + r_2)} - 1$$
$$\hat{r}(3) = \sqrt[3]{(1 + r_1)(1 + r_2)(1 + r_3)} - 1$$

Term Structure Models

Model 1: assumes no drift and that interest rates are normally distributed.
$dr = \sigma dw$

Model 2: adds a positive drift term to Model 1 that can be interpreted as a positive risk premium associated with longer time horizons.
$dr = \lambda dt + \sigma dw$
where:
λ = interest rate drift

Ho-Lee model: generalizes drift to incorporate time-dependency.
$dr = \lambda(t)dt + \sigma dw$

Vasicek model: assumes a mean-reverting process for short-term interest rates.
$dr = k(\theta - r)dt + \sigma dw$
where:
k = speed of reversion
θ = long-run value of the short-term rate
r = current interest rate level

Gauss+ model: more advanced approach than the Vasicek model. Consists of three interacting components: short-term rate, medium-term factor, and long-term factor.

Model 3: assigns a specific parameterization of time-dependent volatility.
$dr = \lambda(t)dt + \sigma e^{-\alpha t}dw$
where:
σ = volatility at $t = 0$, which decreases exponentially to 0 for $\alpha > 0$

Cox-Ingersoll-Ross (CIR) model: mean-reverting model with constant volatility, σ, and basis-point volatility, $\sigma\sqrt{r}$, that increases at a decreasing rate.
$dr = k(\theta - r)dt + \sigma\sqrt{r}dw$

Model 4 (lognormal model): yield volatility, σ, is constant, but basis-point volatility, σr, increases with the level of the short-term rate. There are two lognormal models of importance: (1) lognormal with deterministic drift and (2) lognormal with mean reversion.

Put-Call Parity

$c - p = S - Xe^{-rT}$
where:
c = price of a call
p = price of a put
S = price of the underlying security
r = risk-free rate
T = time left to expiration expressed in years

Volatility Smiles

Currency options: implied volatility is lower for at-the-money options than it is for away-from-the-money options. If the implied volatilities for actual currency options are greater for away-from-the-money than at-the-money options, currency traders must think there is a greater

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chance of extreme price movements than predicted by a lognormal distribution.

Equity options: higher implied volatility for low strike price options. The volatility smirk (half-smile) exhibited by equity options translates into a left-skewed implied distribution of equity price changes. This indicates that traders believe the probability of large down movements in price is greater than large up movements in price, as compared with a lognormal distribution.

CREDIT RISK MEASUREMENT AND MANAGEMENT

Credit Risk Fundamentals

Probability that a creditor will lose money if a borrower fails to honor its financial obligations. This failure can be due to an inability or unwillingness to repay the obligation, or lack of timeliness to repay.

Transactions that generate credit risk: (1) lending, (2) leases, (3) receivables, (4) prepayments, (5) deposits, (6) contingent claims, and (7) derivatives.

Credit-Sensitive Transactions

Transaction parameters: (1) the amount of exposure (measuring losses), (2) credit quality (assessing the risk of losses due to counterparty exposure), and (3) length of exposure (period during which risk exposure exists).

Credit Asset Classifications

Asset classifications help banks categorize loan performance. Categories include: (1) standard (or pass), (2) specially mentioned (or watch), (3) substandard, (4) doubtful, and (5) loss.

Expected Loss (EL)

The expected value of a credit loss is:

$$EL = EA \times PD \times LR$$

where:

Exposure amount (EA): amount of money the lender can lose in the event of a borrower's default.

Probability of default (PD): likelihood that a borrower will default within a specified time horizon.

Loss rate (LR) or *loss given default (LGD)*: amount of creditor loss in the event of a default. In percent terms, it is equal to: 1 – recovery rate.

Unexpected Loss (UL)

Unexpected loss represents the variability of potential losses and can be modeled using the definition of standard deviation.

$$UL = EA \times \sqrt{PD \times \sigma_{LR}^2 + LR^2 \times \sigma_{PD}^2}$$

Capital Adequacy Ratio (CAR)

Banks should have enough capital to match the risk of their assets. This amount is measured with CAR:

$$CAR = \frac{\text{capital}}{\text{risk-weighted assets}} \geq \alpha$$

where:

α = regulatory minimum level (e.g., 8%)

Credit Risk Assessment Approaches

Judgmental (expert) approach: relies on the experience of experts to assess default risk (e.g., 5C Analysis).

Empirical (data-driven) models: uses historical data to look for risk relationships (can apply machine learning).

Financial (market) models: includes structural models (e.g., the Merton model) and reduced-form models (assume default is a random event).

Linear discriminant analysis (LDA) is used to develop credit scoring models (e.g., Altman's Z-score) to classify firms based on credit quality.

Merton Model

Option pricing model where the capital structure of a borrower is viewed as a call option on the assets of the firm (model assumes short-term debt only).

Value of firm equity at *T*:

$$E_T = \max(V_T - D, 0)$$

Value of firm equity today:

$$E_0 = V_\bullet N(d_1) - De^{-rT}N(d_2)$$

The risk-neutral probability of default, $N(-d_2)$, is the probability that shareholders will not exercise their option to repay the loan.

Distance to Default (DD)

Estimates the number of standard deviations that the asset price must fall to lead to a default at *T*:

$$DD = \frac{\ln(V_\bullet) - \ln(D) + (\mu - \sigma_V^2/2)T}{\sigma_V\sqrt{T}}$$

Credit Risk Portfolio Models

Moody's-KMV model: uses both short-term debt and half of long-term debt.

CreditMetrics model: uses a rating transition matrix to account for changes in default rates over time.

CreditRisk+ method: does not account for the capital structure of the borrower and only considers loans in either bankruptcy or a bad credit scenario.

Credit Rating Systems

Through-the-cycle approach: focuses on a long time horizon and covers one full business cycle.

Point-in-time approach: goal is to predict credit quality over a relatively short time horizon.

Country Risk

Sources of country risk: (1) where the country is in the economic growth life cycle, (2) political risks, (3) the legal systems of a country, including both the structure and the efficiency of legal systems, and (4) economic structure, including the level of diversification.

Factors influencing sovereign default risk: (1) a country's level of indebtedness, (2) obligations, such as pension and social service commitments, (3) a country's level of and stability of tax receipts, (4) political risks, and (5) backing from other countries or entities.

Default Probability

The *probability of default*, $Q(t)$, by time *t* is:

$$Q(t) = 1 - e^{-\bar{\lambda}(t) \times t}$$

where:

$\bar{\lambda}(t)$ = average hazard rate

The *hazard rate* (i.e., default intensity) is approximately:

$$\bar{\lambda} = \frac{s(T)}{1 - RR}$$

where:

$s(T)$ = credit spread for maturity *T*

RR = recovery rate

Credit Value at Risk

Credit VaR differs from *market VaR* in that it measures losses that are due specifically to default risk and credit deterioration risk. Credit VaR also typically uses a one-year time horizon.

Credit Derivatives

A *credit derivative* is a contract with payoffs contingent on a specified credit event. Common credit events include: failure to pay or deliver, bankruptcy, and breach of agreement.

Credit Default Swaps (CDSs)

A *CDS* involves a *protection buyer* (insured) paying a periodic CDS spread to a *protection seller* (insurer) for protection against default on a reference obligation. The protection buyer makes these CDS spread premiums until the maturity of the swap or until default, whichever comes first.

Credit Spreads

- The *CDS spread* is the amount required (in percentage) to purchase credit protection.
- The *bond yield spread* is the excess of the corporate bond yield over a risk-free bond.
- The *CDS-bond basis* is:

  CDS spread – bond yield spread

Total Return Swaps (TRSs)

Total return on an asset (bond) is exchanged for a fixed (or variable) payment; total return receiver gets any appreciation (capital gains and cash flows) and pays any depreciation; payments take place whether or not a credit event occurs.

Collateralized Debt Obligations (CDOs)

- *CDOs* pay principal and interest from a collateral pool of debt instruments.
- To create a CDO, the issuer packages a series of debt instruments and splits the package into several classes of securities called *tranches*.
- The largest part of a CDO is typically the senior tranche, which usually carries an AA or AAA credit rating, regardless of the quality of the underlying assets in the pool.
- A *synthetic CDO* is assembled using CDSs rather than debt securities. These CDOs can be valued using either (1) the spread payments approach or (2) the Gaussian copula model.

Clearing Derivatives

- A *derivative* is a contractual agreement between two parties to buy or sell an underlying security or to make a payment in the future.
- Derivatives may be *exchange-traded* (standardized and traded on an exchange), or *over-the-counter (OTC)* (customized). OTC contracts can be centrally cleared, collateralized, or uncollateralized.
- Derivatives are exposed to *counterparty risk*, which is a combination of market risk and credit risk.

Counterparty Risk

The risk that a counterparty is unable or unwilling to live up to its contractual obligations.

Credit exposure: loss that is "conditional" on the counterparty defaulting.

Recovery: measured by the recovery rate, which is the portion of the outstanding claim actually recovered after default.

Wrong-way exposures: exposures that are negatively correlated with the counterparty's credit quality. They increase expected credit losses.

Mark-to-market (MtM): accrual accounting measure that is equal to the sum of the MtM values of all contracts with a given counterparty.

Credit Exposure Metrics

Expected MtM: forward or expected value of a transaction at a given point in the future.

Expected exposure (EE): amount that is expected to be lost (positive MtM only) if the counterparty defaults.

Potential future exposure (PFE): worst exposure that could occur at a given time in the future at a given confidence level.

Expected positive exposure (EPE): average EE through time.

Effective EE: equal to non-decreasing EE.

Effective EPE: average of effective EE.

Maximum PFE: highest PFE value over a stated time frame.

Credit Mitigation Techniques

Netting: a legally binding agreement that enables counterparties with multiple derivatives contracts to net their obligations (e.g., Party A owes Party B \$50 million; Party B owes Party A \$40 million, so Party A pays a net \$10 million to Party B).

Collateralization: if the value of the derivatives contracts is above a stated threshold, collateral must equal the difference between the value of the contracts and the threshold level.

Modeling Collateral

Certain parameters impact the effectiveness of collateral in lessening credit exposure. These parameters are as follows:

Margin period of risk: the time between the call for collateral and its receipt.

Threshold: an exposure level below which collateral is not called. It represents an amount of uncollateralized exposure.

Minimum transfer amount: the minimum quantity or block in which collateral may be transferred. Quantities below this amount represent uncollateralized exposure.

Initial margin: an amount posted independently of any subsequent collateralization. This is also referred to as the independent amount.

Rounding: the process by which a collateral call amount will be adjusted (rounded) to a certain increment.

Central Counterparties (CCPs)

CCPs provide clearing services for financial transactions between member firms. They use netting to reduce the number of transactions.

CCP advantages: transparency, multilateral offset, loss mutualization, legal and operational efficiency, liquidity, and default management.

CCP disadvantages: moral hazard, adverse selection, bifurcations, and procyclicality.

Credit Value Adjustment (CVA)

Expected value or price of counterparty credit risk. The CVA represents a cost to the counterparty that bears a greater propensity to default. It should account for the counterparty's default probability, the transaction in question, netting, collateral, and hedging.

$$CVA = -LGD \times \sum_{i=1}^{m} \times EPE(t_i) \times PD(t_{i-1}, t_i)$$

where:

EPE = discounted expected positive exposure

Debt Value Adjustment (DVA)

Financial institutions should incorporate the value of their option to default to a counterparty, known as the DVA, by using the bilateral CVA (BCVA) equation. Unlike the CVA, the DVA incorporates expected negative exposure (ENE). CVA and DVA are mirror images of each other.

Wrong-Way Risk vs. Right-Way Risk

Wrong-way risk (WWR): increases counterparty risk (increases CVA and decreases DVA).

Right-way risk (RWR): decreases counterparty risk (decreases CVA and increases DVA).

WWR Modeling Methods

Intensity approach (i.e., hazard rate approach): uses stochastic process for credit spreads.

Structural approach: considers combination of default and exposure distributions.

Parametric approach: evaluates link between portfolio values and credit spreads.

Jump approach: considers that foreign exchange rates may jump upon default.

Securitization

Transforms the illiquid assets of a financial institution into a package of asset-backed securities (ABSs) or mortgage-backed securities (MBSs). A third party uses credit enhancements, liquidity enhancements, and structuring to issue securities backed by the pooled cash flows (of the same underlying assets).

- *Credit enhancements* include overcollateralization, subordinating note classes, margin step-up, and excess spread.
- *The first-loss piece (equity piece)* absorbs initial losses and is often held by the originator.

ABS/MBS Performance Tools

Auto loans: loss curves, absolute prepayment speed.

Credit card debt: delinquency ratio, default ratio, monthly payment rate.

Mortgages: debt service coverage ratio, weighted average coupon, weighted average maturity, weighted average life, single monthly mortality, constant prepayment rate, Public Securities Association.

OPERATIONAL RISK AND RESILIENCE

Operational Risk Management

The Basel Committee on Banking Supervision defines operational risk as "the risk of loss resulting from inadequate or failed internal processes, people and systems or from external events." The four risk management steps to be taken are: (1) risk identification, (2) risk assessment, (3) risk mitigation, and (4) risk monitoring/reporting.

Operational Risk Event Types

- Internal Fraud.
- External Fraud.
- Employment Practices and Workplace Safety.
- Clients, Products, and Business Practices.
- Damage to Physical Assets.
- Business Disruption and System Failures.
- Execution, Delivery, and Process Management.

Three Lines of Defense

Lines of defense to control operational risks include (1) risk management in each business unit, (2) corporate operational risk management function (CORF), and (3) independent reviews of operational risks and risk management.

Risk Identification

For risk identification, the two approaches to consider are top down and bottom up. Common top-down approaches: (1) analyzing exposures and vulnerabilities, (2) risk wheel, and (3) analyzing emerging risks. Common bottom-up approaches: (1) event and loss data analysis, (2) risk and control self-assessment (RCSA), and (3) process mapping.

Risk Assessment

Operational risk management and measurement makes use of historical losses and events. Qualitative risk assessment includes: RCSAs, which assess probability and severity of operational risks, and key risk indicators (KRIs), which indicate bank exposure level at a specific point in time. Quantitative risk assessment includes: fault tree analysis (FTA), which break failure scenarios into external and internal conditions for event occurrence, and root-cause analysis, which examines the causes of incidents (e.g., bow tie approach).

Operational Resilience

The main steps for operational resilience are (1) determine important business services, (2) establish impact tolerances, (3) map important business services, (4) test vulnerabilities, (5) review lessons learned, (6) ensure communication plans are ready, and (7) annual self-assessment.

Risk Mitigation

Risk mitigation identifies the measures a firm can undertake to minimize operational risk to acceptable levels. Operational risk cannot be eliminated, but it can be reduced. Reducing operational risk can generally be done by (1) establishing strong controls, (2) buying insurance, or (3) reducing certain activities. For risk mitigation with internal controls, the main types are: (1) preventive, (2) detective, (3) corrective, and (4) directive. Approaches for reducing the impact of operational risk events include: (1) contingency planning, (2) resilience measurement, (3) crisis management, and (4) communication.

Risk Monitoring/Reporting

Risk monitoring provides organizational leaders with a view of the organization risk profile, assurance of operating within risk appetite limits, and the tools to evaluate the effectiveness of the risk management framework. An internal operational risk management report contains: (1) prioritized risks and outlook, (2) heatmap and risk register, (3) risk appetite metrics, (4) key risk indicators (KRIs) and issue monitoring, (5) risk events and near misses, (6) action plans and remediation, and (7) horizon scanning.

Integrated Risk Management

The enterprise risk management (ERM) framework includes the tools and methods that an organization uses to manage the different types of risks that can impact the achievement of business objectives. ERM and related priorities are guided by:

- *Risk governance*: establishes the committees responsible for reporting and decision-making.
- *Risk culture*: relates to the behaviors and values associated with managing risk.
- *Risk appetite*: helps firms define priorities and the levels of risk exposure they are willing to tolerate.

Cyber Resilience

An effective cyber resilience framework consists of the following elements: (1) identify, (2) protect, (3) detect, (4) respond, and (5) recover.

Money Laundering and the Financing of Terrorism (ML/FT)

Banks must determine which customers pose a low or high risk of ML/FT using customer due diligence (CDD). Lines of defense to mitigate ML/FT risks include (1) business units (e.g., customer facing activities), (2) the chief ML/FT officer, and (3) internal and external audits.

Third-Party Risk Management

An effective program to manage outsourcing risk should include (1) risk assessments, (2) due diligence in selecting service providers, (3) contract provisions, (4) incentive compensation review, (5) oversight and monitoring of service providers, and (6) business continuity and contingency plans.

Model Risk

Model risk raises the possibility of (negative) outcomes resulting from inaccurate model outputs. It can arise in two ways: (1) model has significant errors and produces faulty outputs, and (2) model is used out of context or is not used properly for its intended purposes.

Risk-Adjusted Return on Capital

The RAROC measure is essential to successful integrated risk management. Its main function is to relate the return on capital to the riskiness of firm investments. The RAROC is risk-adjusted return divided by risk-adjusted capital (i.e., economic capital).

$$\text{RAROC} = \frac{\left(\begin{array}{l}\text{revenues} - \text{costs} - \text{EL} - \text{taxes} + \\ \text{return on economic capital} \pm \text{transfers}\end{array}\right)}{\text{economic capital}}$$

An adjusted RAROC (ARAROC) measure better aligns the risk of the business with the risk of the firm's equity.

$$\text{Adjusted RAROC} = \text{RAROC} - \beta_E(R_M - R_F)$$

Capital Plan Rule

- Mandates that bank holding companies develop a capital plan and evaluate capital adequacy.
- Capital adequacy process includes: risk management foundation, resource and loss estimation methods, impact on capital adequacy, capital planning and internal controls policies, and governance oversight.

Basel II: Three Pillars

Pillar 1: Minimum capital requirements. Banks should maintain a minimum level of capital to cover credit, market, and operational risks.

Pillar 2: Supervisory review process. Banks should assess the adequacy of capital relative to risk, and supervisors should review and take corrective action if problems occur.

Pillar 3: Market discipline. Risks should be adequately disclosed in order to allow market participants to assess a bank's risk profile and the adequacy of its capital.

Market Risk Capital Requirements

Standardized method: determines capital charges associated with various market risk exposures (equity risk, interest rate risk, foreign exchange risk, commodity risk, and option risk). The market risk capital charge for each market risk is computed as 8% of its market-risky assets.

Internal models approach: allows a bank to use its own risk management systems to determine its market risk capital charge. The market risk charge is the higher of (1) the previous day's VaR or (2) the average VaR over the last 60 business days adjusted by a multiplicative factor (subject to a floor of 3).

Backtesting VaR

An exception occurs if the day's change in value exceeded the VaR estimate of the previous day. When backtesting VaR, the number of exceptions is determined for a 250-day testing period. Based on the number of exceptions, the bank's exposure is categorized into one of three zones and VaR is scaled up by the appropriate multiplier (subject to a floor of 3).

- Green zone: 0–4 exceptions, increase in exposure multiplier is 0.
- Yellow zone: 5–9 exceptions, exposure multiplier increases between 0.4 and 0.85.
- Red zone: Greater than or equal to 10 exceptions, multiplier increases by 1.

Credit Risk Capital Requirements

The *standardized approach* incorporates risk weights based on external credit rating assessments. The amount of capital that a bank must hold is specific to the risk of credit-risky assets, the type of institution the claim is written on, and the maturity of those assets.

The *internal ratings-based (IRB)* approaches (foundation and advanced) use a bank's own internal estimates of creditworthiness to determine the risk weightings in the capital calculation.

- *Foundation approach:* bank estimates PD.
- *Advanced approach:* bank estimates not only PD, but also LGD, exposure at default (EAD), and effective maturity (M).

Operational Risk Capital Requirements

Basic indicator approach: measures the capital charge on a firm-wide basis. Banks will hold capital for operational risk equal to a fixed percentage of the bank's average annual gross income over the prior three years. The Basel Committee has proposed a fixed percentage equal to 15%.

Standardized approach: allows banks to divide activities along standardized business lines. Within each business line, gross income will be multiplied by a fixed beta factor. The capital charge for operational risk is the sum of each business line's charges.

Advanced measurement approach (AMA): If a bank can meet more rigorous supervisory standards, it may use the AMA for operational risk capital calculations. The capital charge for AMA is calculated as the bank's operational value at risk with a 1-year horizon and a 99.9% confidence level. Having insurance can reduce this capital charge by as much as 20%.

Solvency II

Establishes capital requirements for the operational, investment, and underwriting risks of insurance companies.

- Specifies minimum capital requirements (MCR) and solvency capital requirements (SCR).
- *Standardized approach:* Intended for less sophisticated insurance firms; captures the risk profile of the average insurance firm.
- *Internal models approach:* Similar to the IRB approach under Basel II. A VaR is calculated with a one-year time horizon and a 99.5% confidence level.

Stressed Value at Risk (SVaR)

SVaR is calculated by combining current portfolio performance data with the firm's historical data from a significantly financial stressed period in the same portfolio:

$$\max(\text{SVaR}_{t-1}, \text{multiplicative factor} \times \text{SVaR}_{avg})$$

Basel III Changes

- Raise capital standards (both quality and quantity).
- Strengthen risk coverage of capital framework.
- Require leverage ratio to supplement capital requirements.
- Promote countercyclical buffers during financial shocks.
- Institute policies to address systemic risk and interconnectedness.
- Institute global liquidity standard (liquidity, funding, and monitoring metrics).

Liquidity Coverage Ratio (LCR)

Goal: ensure banks have adequate, high-quality liquid assets to survive short-term stress scenario.

$$\text{LCR} = (\text{stock of high-quality liquid assets} / \text{total net cash outflows over next 30 calendar days}) \geq 100$$

Net Stable Funding Ratio (NSFR)

Goal: protect banks over a longer time horizon than LCR.

$$\text{NSFR} = (\text{available amount of stable funding} / \text{required amount of stable funding}) \geq 100$$

Standardized Measurement Approach (SMA)

The SMA for operational risk includes both a business indicator (BI) component accounting for operational risk exposure and an internal loss multiplier (and loss component) accounting for operational losses unique to an individual bank. The BI component impact will vary depending on where the bank is classified from buckets 1–5.

LIQUIDITY AND TREASURY RISK MEASUREMENT AND MANAGEMENT

Liquidity Risk

The lack of a market for a security to prevent it from being traded quickly enough to prevent or minimize a loss.

Trading liquidity risk: risk that the act of buying or selling an asset will result in an adverse price move.

Funding liquidity risk: results when a borrower's credit position is either deteriorating or perceived by market participants to be deteriorating.

Bid-Offer (or Bid-Ask) Spread

$$s = \frac{\text{offer price} - \text{bid price}}{\text{mid-market price}}$$

Liquidity-Adjusted VaR (LVaR)

$$\text{LVaR} = \text{VaR} + \text{cost of liquidation}$$

where:

$$\text{cost of liquidation} = \sum_{i=1}^{n} \frac{s_i \alpha_i}{2}$$

α = mid-market value

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Leverage Ratio

A firm's leverage ratio is equal to its assets divided by equity:

$$L = \frac{A}{E} = \frac{(E + D)}{E} = 1 + \frac{D}{E}$$

Leverage Effect

Return on equity (ROE) is higher as leverage increases, as long as the firm's return on assets (ROA) exceeds the cost of borrowing funds. The leverage effect can be expressed as:

ROE = (leverage ratio × ROA) – [(leverage ratio – 1) × cost of debt]

Transactions Cost

The expected transactions cost is:

$P \times (s/2)$

The 99% confidence interval:

$+/- P \times ½(s + 2.33\sigma_S)$

where:

P = estimate of the next day asset midprice

s = bid-ask spread

$½(s + 2.33\sigma_S)$ = 99% spread risk factor

Early Warning Indicators (EWIs)

EWIs are changes in key metrics that could signal a pending liquidity problem. They should contain both internal and external measures, be leading indicators and granular, provide warning of any potential worsening of a bank's financial position, identify whether the bank has sufficient liquid resources to handle a stress scenario, and consider different time horizons.

Net Liquidity Position

supplies of liquidity – demands for liquidity

where:

supplies of liquidity = incoming deposits (inflows) + customer loan repayments + asset sales + revenue from nondeposit services + money market borrowings

demands for liquidity = deposit withdrawals (outflows) + borrowing repayments + dividend payments + loan requests + other operating expenses

Liquidity Requirements

Methods for estimating liquidity needs include (1) the sources and uses of funds approach, (2) the structure of funds approach, (3) the liquidity indicator approach, and (4) the market signals (discipline) approach.

Intraday Liquidity

Uses: outgoing wire transfers, payment clearing and settlement (PCS) systems, funding of foreign accounts, collateral pledging, and asset purchases/funding.

Sources: cash balances, payment inflows, intraday credit, liquid assets, and overnight borrowings.

Deterministic and Stochastic Cash Flows

In terms of time, cash flows can be deterministic (occurring at known times) or stochastic (occurring randomly). Similarly, cash flows with known amounts are deterministic (i.e., fixed), while cash flows with unknown amounts are stochastic (e.g., credit related, behavioral).

Contingent Liquidity

Comprised of the (very high quality) liquid assets and credit facilities that are meant to satisfy general liabilities in stressed situations. Contingent liquidity is estimated using the liquid asset buffer, which includes assets that typically have little or no credit and market risk, are easy to value, and are actively traded. The stressed liquidity asset buffer is estimated as:

(normal) liquidity asset buffer – stressed cash outflows + stressed cash inflows

Contingency Funding Plans (CFPs)

Design considerations of CFPs include:

- Alignment to business and risk profiles.
- Integration with broader risk management frameworks.
- Operational, actionable, and flexible playbooks.
- Inclusive of appropriate stakeholder groups.
- Communication plan support.

Methods for Pricing Deposits

Cost-plus pricing: prices the deposit service such that the amount covers the direct and overhead costs associated with providing the service, as well as a profit margin.

Marginal cost pricing: compares the cost of raising additional funds with the yield the financial institution earns on the assets with which it invests the additional funds.

Conditional pricing: bases deposit fees on a condition, such as a minimum balance to be maintained in an account.

Available Funds Gap (AFG)

The difference between the current and projected inflows and outflows of bank funds.

AFG = current and projected loans and other investments – current and expected deposit inflows and other available funds

Repurchase Agreements (Repos)

- Bilateral contracts where one party sells a security at a specified price with a commitment to buy back the security at a future date at a higher price.
- From the perspective of the *borrower*: repos offer relatively cheap sources of short-term funds.
- From the perspective of the *lender*: reverse repos are used for either investing or financing purposes.

Liquidity Transfer Pricing (LTP)

LTP is the process of attributing the costs, benefits, and risks associated with liquidity to appropriate business units of the bank. Approaches used in LTP include zero cost, pooled average cost, separate average cost, and marginal cost. LTP best practice is to use the marginal cost approach, which incorporates actual market costs of funding to calculate liquidity spread.

Covered Interest Parity (CIP)

The cross-currency swap basis (*b*) is the amount by which the USD interest rate must be adjusted so that CIP holds (assuming F – S is too wide).

$$F - S = S\left(\frac{1 + r_{USD} + b}{1 + r_{FC}}\right) - S$$

Factors that can cause deviations in CIP and cross-currency swap basis include (1) the lack of market liquidity and (2) increases in the risk of the underlying instruments that gives rise to risk premiums.

Net Interest Margin (NIM)

Measure of bank performance: (interest income – interest expense) / bank assets that earn income

Interest-Sensitive (IS) Gap

Measure of interest rate risk: interest-sensitive assets – interest-sensitive liabilities

Duration Gap

Measure of the market (price) risk on assets relative to the market (price) risk on liabilities. The *leverage-adjusted duration gap* measures the effect of interest rate changes on the balance sheet.

$$D_A - D_L \times \frac{\text{total liabilities}}{\text{total assets}}$$

Illiquid Asset Return Biases

Biases that impact reported illiquid asset returns:

- *Survivorship bias*: Poor performing funds often quit reporting results, ultimately fail, or never begin reporting returns because performance is weak.
- *Selection bias*: Asset values and returns tend to be reported when they are high.
- *Infrequent trading*: Betas, volatilities, and correlations are too low when they are computed using the reported returns of infrequently traded assets.

RISK MANAGEMENT AND INVESTMENT MANAGEMENT

Factor Risks

Represent exposures to bad times; must be compensated for with risk premiums. Factor risk principles:

- It is not exposure to the specific asset that matters, rather the exposure to the underlying risk factors.
- Assets represent bundles of factors, and assets' risk premiums reflect these risk factors.
- Investors have different optimal exposures to risk factors, including volatility.

Fama-French Model

Explains asset returns based on:

- Traditional capital asset pricing model (CAPM) market risk factor.
- Factor that captures size effect (SMB or small cap minus big cap).
- Factor that captures value/growth effect (HML or high book-to-market value minus low book-to-market value).

Momentum effect: long winners and short losers (WML or winners minus losers). This strategy has outperformed both size and value/growth effects; however, it is subject to crashes.

Fundamental Law of Active Management

Tradeoff between required degree of forecasting accuracy (information coefficient [IC]) and number of investment bets placed (breadth [BR]). The *information ratio* (IR) is:

$IR \approx IC \times \sqrt{BR}$

Portfolio Construction Techniques

- *Screens* simply choose assets by ranking alpha.
- *Stratification* chooses stocks based on screens; includes assets from all asset classes.
- *Linear programming* attempts to construct a portfolio that closely resembles the benchmark.
- *Quadratic programming* explicitly considers alpha, risk, and transaction costs.

PART II SCHWESER'S QuickSheet

CRITICAL CONCEPTS FOR THE 2025 FRM® EXAM

MARKET RISK MEASUREMENT AND MANAGEMENT

Value at Risk (VaR)

VaR for a given confidence level occurs at the cutoff point that separates the tail losses from the remaining distribution.

Historical simulation approach: order return observations and find the observation that corresponds to the VaR loss level.

Parametric estimation approach: assumes a distribution for the underlying observations.

- Normal distribution assumption:

$$\text{VaR} = (-\mu_r + \sigma_r \times z_\alpha)$$

- Lognormal distribution assumption:

$$\text{VaR} = (1 - e^{\mu_r - \sigma_r \times z_\alpha})$$

Expected Shortfall (ES)

Estimates tail loss by averaging VaRs for increasing confidence levels in the tail.

Weighted Historical Simulation Approaches

- *Age-weighted*: adjusts the most recent (distant) observations to be more (less) heavily weighted.
- *Volatility-weighted*: replaces historic returns with volatility-adjusted returns; actual procedure of estimating VaR is unchanged.
- *Correlation-weighted*: updates the variance-covariance matrix between assets in the portfolio.
- Filtered historical simulation: relies on bootstrapping of standardized returns based on volatility forecasts; able to capture conditional volatility, volatility clustering, and/or data asymmetry.

Peaks-Over-Threshold (POT)

Application of extreme value theory (EVT) to the distribution of excess losses over a high threshold. The POT approach can be used to compute VaR. From estimates of VaR, we can derive ES.

Backtesting VaR

- Compares the number of instances when losses exceed the VaR level (exceptions or exceedances) with the number predicted by the model at the chosen level of confidence.
- *Failure rate*: number of exceptions divided by number of observations.
- The Basel Committee requires backtesting at the 99% confidence level over one year; establishes zones for the number of exceptions with corresponding penalties (increases in the capital multiplier).

VaR Mapping

Mapping involves finding common risk factors among positions in a given portfolio. It may be difficult and time consuming to manage the risk of each individual position. One can evaluate the value of portfolio positions by mapping them onto common risk factors.

Mean Reversion

- Implies that over time variables or returns regress back to the mean or average return.
- Mean reversion rate, a, is expressed as:

$$S_t - S_{t-1} = a(\mu - S_{t-1})$$

- The β coefficient of a regression is equal to the negative of the mean reversion rate.

Autocorrelation

- Measures the degree that a variable's current value is correlated to past values.
- Has the exact opposite properties of mean reversion.
- The sum of the mean reversion rate and the one-period autocorrelation rate will always equal one.

Correlation Swap

- Used to trade a fixed correlation between two or more assets with a realized correlation.
- Realized correlation for a portfolio of n assets:

$$\rho_{\text{realized}} = \frac{2}{n^2 - n} \sum_{i>j} \rho_{i,j}$$

- Payoff for correlation swap buyer:
  notional amount $\times (\rho_{\text{realized}} - \rho_{\text{fixed}})$

Gaussian Copula Model

- Indirectly defines a correlation relationship between two variables.
- Maps the marginal distribution of each variable to a standard normal distribution (done on percentile-to-percentile basis).
- The new joint distribution is a multivariate standard normal distribution.
- A Gaussian default time copula can be used for measuring the joint probability of default between two assets.

Regression-Based Hedge

$$F^R = F^N \times \left(\frac{DV01^N}{DV01^R}\right) \times \beta$$

where:
F^R = face amount of hedging instrument
F^N = face amount of initial position

Bond Valuation Using Binomial Tree

Using backward induction, the value of a bond at a given node in a binomial tree is the average of the present values of the two possible values from the next period. The appropriate discount rate is the forward rate associated with the node under analysis.

There are three basic steps to valuing an option on a fixed-income instrument using a binomial tree.

Step 1: Price the bond value at each node using the projected interest rates.

Step 2: Calculate the intrinsic value of the derivative at each node at maturity.

Step 3: Calculate the expected discounted value of the derivative at each node using the risk-neutral probabilities and work backward through the tree.

Interest Rate Expectations

Expectations play an important role in determining the shape of the yield curve and can be illustrated by examining yield curves that are flat, upward-sloping, and downward-sloping. If expected 1-year spot rates for the next three years are r_1, r_2, and r_3, then the 2-year and 3-year spot rates are computed as follows.

$$\hat{r}(2) = \sqrt{(1 + r_1)(1 + r_2)} - 1$$

$$\hat{r}(3) = \sqrt[3]{(1 + r_1)(1 + r_2)(1 + r_3)} - 1$$

Term Structure Models

Model 1: assumes no drift and that interest rates are normally distributed.

$dr = \sigma dw$

Model 2: adds a positive drift term to Model 1 that can be interpreted as a positive risk premium associated with longer time horizons.

$dr = \lambda dt + \sigma dw$

where:

λ = interest rate drift

Ho-Lee model: generalizes drift to incorporate time-dependency.

$dr = \lambda(t)dt + \sigma dw$

Vasicek model: assumes a mean-reverting process for short-term interest rates.

$dr = k(\theta - r)dt + \sigma dw$

where:

k = speed of reversion

θ = long-run value of the short-term rate

r = current interest rate level

Gauss+ model: more advanced approach than the Vasicek model. Consists of three interacting components: short-term rate, medium-term factor, and long-term factor.

Model 3: assigns a specific parameterization of time-dependent volatility.

$dr = \lambda(t)dt + \sigma e^{-\alpha t}dw$

where:

σ = volatility at $t = 0$, which decreases exponentially to 0 for $\alpha > 0$

Cox-Ingersoll-Ross (CIR) model: mean-reverting model with constant volatility, σ, and basis-point volatility, $\sigma\sqrt{r}$, that increases at a decreasing rate.

$dr = k(\theta - r)dt + \sigma\sqrt{r}dw$

Model 4 (lognormal model): yield volatility, σ, is constant, but basis-point volatility, σr, increases with the level of the short-term rate. There are two lognormal models of importance: (1) lognormal with deterministic drift and (2) lognormal with mean reversion.

Put-Call Parity

$c - p = S - Xe^{-rT}$

where:

c = price of a call

p = price of a put

S = price of the underlying security

r = risk-free rate

T = time left to expiration expressed in years

Volatility Smiles

Currency options: implied volatility is lower for at-the-money options than it is for away-from-the-money options. If the implied volatilities for actual currency options are greater for away-from-the-money than at-the-money options, currency traders must think there is a greater

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